30/05/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

PROJECT PHOENIX FOR RE-NEET-2026

MM: 720 **CST - 03** **Time: 180 Mins.**

Answers

1. (2)	37. (3)	73. (2)	109. (4)	145. (2)
2. (3)	38. (2)	74. (3)	110. (4)	146. (3)
3. (4)	39. (2)	75. (3)	111. (2)	147. (2)
4. (1)	40. (2)	76. (2)	112. (3)	148. (4)
5. (1)	41. (1)	77. (4)	113. (4)	149. (3)
6. (3)	42. (1)	78. (4)	114. (3)	150. (2)
7. (4)	43. (3)	79. (1)	115. (4)	151. (3)
8. (2)	44. (2)	80. (3)	116. (1)	152. (1)
9. (1)	45. (1)	81. (2)	117. (1)	153. (3)
10. (3)	46. (1)	82. (4)	118. (2)	154. (4)
11. (2)	47. (3)	83. (1)	119. (3)	155. (3)
12. (4)	48. (1)	84. (4)	120. (3)	156. (2)
13. (2)	49. (1)	85. (2)	121. (1)	157. (1)
14. (1)	50. (3)	86. (2)	122. (3)	158. (3)
15. (1)	51. (4)	87. (4)	123. (4)	159. (4)
16. (3)	52. (4)	88. (4)	124. (4)	160. (3)
17. (1)	53. (3)	89. (4)	125. (2)	161. (3)
18. (4)	54. (1)	90. (3)	126. (4)	162. (2)
19. (4)	55. (2)	91. (2)	127. (4)	163. (2)
20. (3)	56. (1)	92. (4)	128. (4)	164. (1)
21. (2)	57. (1)	93. (4)	129. (1)	165. (4)
22. (1)	58. (2)	94. (4)	130. (1)	166. (2)
23. (3)	59. (4)	95. (2)	131. (2)	167. (3)
24. (4)	60. (3)	96. (1)	132. (4)	168. (1)
25. (4)	61. (3)	97. (1)	133. (2)	169. (1)
26. (4)	62. (4)	98. (3)	134. (3)	170. (2)
27. (2)	63. (3)	99. (3)	135. (3)	171. (1)
28. (3)	64. (1)	100. (1)	136. (1)	172. (4)
29. (1)	65. (1)	101. (1)	137. (3)	173. (3)
30. (3)	66. (1)	102. (3)	138. (3)	174. (4)
31. (3)	67. (4)	103. (3)	139. (1)	175. (2)
32. (2)	68. (1)	104. (3)	140. (4)	176. (3)
33. (2)	69. (2)	105. (4)	141. (3)	177. (1)
34. (4)	70. (2)	106. (4)	142. (1)	178. (3)
35. (4)	71. (4)	107. (1)	143. (3)	179. (1)
36. (2)	72. (1)	108. (3)	144. (4)	180. (3)

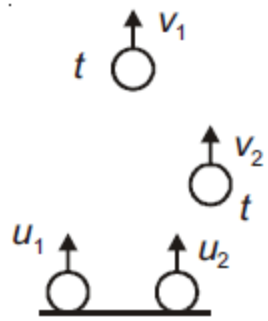
Hints and Solutions

PHYSICS

(1) **Answer :** (2)
Solution:
$x = at^3 + bt^2 + ct + d$
$V = 3at^2 + 2bt + c$
$a = 6at + 2b$
at $t = 3$ s
$a = 18a + 2b = 2(9a + b)$

(2) **Answer :** (3)
Solution:
$\frac{dy}{dt} = cyx^2$
$\frac{dx}{dt} = cxy^2$
$\frac{dy}{dx} = \frac{yx^2}{xy^2} = \frac{x}{y}$
⇒ $ydy = xdx$
⇒ $\frac{y^2}{2} = \frac{x^2}{2} + \text{constant}$
⇒ $y^2 = x^2 +$ constant

(3) **Answer :** (4)
Solution:



$v_1 = u_1 - gt$
$v_2 = u_2 - gt$
$v_1 - v_2 = u_1 - u_2$
$\boxed{v_{12} = u_1 - u_2}$
Time independent

(4) **Answer :** (1)
Solution: $T = \frac{2u\sin\theta}{g} = \frac{2u_y}{g} = \frac{2\beta}{g}$

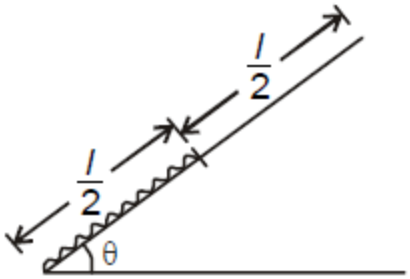
(5) **Answer :** (1)
Solution:
True value $= \frac{1.45+1.56+1.54+1.44+1.54+1.53}{6}$
True value = 1.51
Mean absolute error
$= \frac{(|0.06|+|0.05|+|0.03|+|0.07|+|0.03|+|0.02|)}{6}$
= 0.04

(6) **Answer :** (3)
Solution:
$p_0 = \frac{4}{\sqrt{2}}\hat{i}$ kg m/s and

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$p_f = \frac{4}{\sqrt{2}}\hat{j}$ kg m/s $\Rightarrow \Delta p = \frac{4}{\sqrt{2}}\left(\hat{j} - \hat{i}\right)$

(7) **Answer :** (4)
Solution:



For upperhalf

$v^2 = u^2 + 2a\frac{l}{2}$

$\Rightarrow\ v^2 = 0 + 2\left(g\sin\theta\right)\frac{l}{2}$

$\Rightarrow\ v^2 = gl\sin\theta$ ...(i)

and $0 = u'^2 + 2g\left(\sin\theta - \mu\cos\theta\right)\frac{l}{2}$

$\Rightarrow\ -gl\sin\theta = 2g\left(\sin\theta - \mu\cos\theta\right)\frac{l}{2}$

$\boxed{\mu = 2\tan\theta}$

(8) **Answer :** (2)
Hint:

$dW = \vec{F}\,.\,\vec{dr}$

Solution:

$\vec{F} = 2x\hat{i} + 3\hat{j} + 3z^2\hat{k}$

$\vec{dr} = dx\hat{i} + dy\hat{j} + dz\,\hat{k}$

$dW = \left(2x\hat{i} + 3\hat{j} + 3z^2\hat{k}\right).\left(dx\hat{i} + dy\hat{j} + d_2\hat{k}\right)$

$W = \left[x^2\right]^3{}_2 + [y]^2{}_3 + \left[z^3\right]^1{}_0$

W = 5 – 3 +1

W = + 3 J

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(9) **Answer :** (1)
Solution:
Loss in P ·E = Δ (K·E)*r*

(10) **Answer :** (3)
Solution:
Reading of venturimeter calculated with the help of Bernoulli's equation $P_0 + \frac{1}{2}\rho v^2 + \rho gh$ = constant and principle of continuity.

(11) **Answer :** (2)
Hint:

$v_e = \sqrt{\frac{2GM}{R}}$

Solution:

$v_e = \sqrt{\frac{2GM}{R}} = R\sqrt{\frac{8}{3}\pi\rho G}$

∴ $v_e \propto R$ [ρr = constant]

here $\frac{v_{e1}}{v_{e2}} = \frac{R_1}{R_2} = \frac{1}{2}\ \Rightarrow\ v_{e,\text{ planet}} = 2v_{e,\text{ earth}}$

= 22.4 km/s

(12) **Answer :** (4)
Solution:
the bulk modulus measures a substance's resistance to uniform compression, which applies to all three states of matter.

The shear modulus measures a material's response to shear stress, and it is relevant for solids, not liquids, as liquids cannot sustain shear stress.
Young's modulus measures the tensile elasticity of a solid material.

(13) **Answer :** (2)
Solution:
$R_{\text{th}} = \frac{9}{K_y A} + \frac{3}{K_x A}$
$= \frac{9}{K_y A} + \frac{3}{3K_x A}\ \ [\because K_x = 3K_y]$
$= \frac{10}{K_y A}$
$i_{\text{th}} = \frac{200-100}{R_{\text{th}}} = \frac{100 \times K_y A}{10} = 10K_y A$

now, $10K_y A = \frac{K_y A(200^\circ - T)}{9}$
$200^\circ - T = 90^\circ$
$T = 110^\circ\text{C}$

(14) **Answer :** (1)
Solution:
$V_{\text{av}} = \sqrt{\frac{8RT}{\pi M}}$
$v = \sqrt{\frac{8R(200)}{\pi M}}$
$v^1 = \sqrt{\frac{8R(400)}{\pi M}}$
$\frac{\text{v}^1 - \text{v}}{\text{v}} \times 100\% = \left[\frac{\sqrt{400} - \sqrt{200}}{\sqrt{200}}\right] \times 100\%$
$= \left[\sqrt{\frac{4}{2}} - 1\right] \times 100\%$
$= (0.414) \times 100\% = 41.4\%$

(15) **Answer :** (1)
Solution:
$x = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{600} = \frac{1}{2}$

(16) **Answer :** (3)
Solution:
$W_{\text{isob}} > W_{\text{isot}} > W_{\text{adia}}$ from area under *P-V* graph.

(17) **Answer :** (1)
Solution:
Since wave is moving towards negative *x* direction
∴ replace *x* with (*x* + 2*t*)

(18) **Answer :** (4)
Solution:
$\frac{20}{\sqrt{2}} = 20\sin[4\pi(0) + \theta]$
$\sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$ radian
Total phase $= \frac{4\pi}{15} \times (1.5) + \frac{\pi}{4}$
$= \frac{4\pi}{10} + \frac{\pi}{4} = \frac{13}{20}\pi$ rad

(19) **Answer :** (4)
Solution:
$\frac{1}{2}m\omega^2 A^2 = 8 \times 10^{-3}$
$\frac{1}{2} \times 0.1 \times \omega^2 \times (0.1)^2 = 8 \times 10^{-3}$
ω = ±4
∴ $y = A\sin(\omega t + \phi) = 0.1\sin\left(4t + \frac{\pi}{4}\right)$

(20) **Answer :** (3)
Solution:
$\frac{\lambda}{2} = 29.5 - 10.5 = 19\text{ cm}$

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So, for 3rd resonance 29.5 + 19 = 48.5 cm

(21) **Answer :** (2)
Solution:
$W = \frac{1}{2}K\left(x_2^2 - x_1^2\right)$
= $\frac{1}{2} \times 800(225 - 25) \times 10^{-4}$
= 8 J

(22) **Answer :** (1)
Solution:
$v = \sqrt{2gh}$
$= \sqrt{2 \times 10 \times 5} = 10\ \mathrm{m/s}$

(23) **Answer :** (3)
Solution:
As the conducting surface is an equipotential surface in electrostatic equilibrium, therefore electrostatic field lines are normal to the conducting surface.

(24) **Answer :** (4)
Solution:
Effective capacitance of circuit = $\frac{5C}{2}$
$\frac{5C}{2} \times 6 = 1.5$
$C = \frac{3}{30} = 0.1\ \mu\mathrm{F}$

(25) **Answer :** (4)
Solution:
$q = ne$
$n = \frac{1.6 \times 10^{-18}}{1.6 \times 10^{-19}}$ = 10

(26) **Answer :** (4)
Solution:
$\tau = PE\sin\theta$
$= 10^{-8} \times 2 \times \frac{\sqrt{3}}{2}$
$= \sqrt{3} \times 10^{-8}\ \mathrm{N-m}$

(27) **Answer :** (2)
Solution:
Particle at rest experiences electromagnetic force then, electric field must not be zero and magnetic field may or may not be zero.

(28) **Answer :** (3)
Solution:
15 V – 9 V ⇒ 3 V = iR_1
Similarly,
9 V – 6 V ⇒ 3 V = iR_2
6 V – 0 V ⇒ 6 V = iR_3
$R_1 : R_2 : R_3$
2 : 1 : 2

(29) **Answer :** (1)
Solution:
Due to rise in current brightness will increase.

(30) **Answer :** (3)
Solution:
$\mu_0 \rightarrow \left[\mathrm{MLT^{-2}A^{-2}}\right]$
$B \rightarrow \left[\mathrm{MT^{-2}A^{-1}}\right]$
$\mathrm{M} \rightarrow \left[\mathrm{L^2A}\right]$
$C \rightarrow \left[\mathrm{ML^2T^{-2}}\right]$

(31) **Answer :** (3)

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Solution:

$U_1 = \frac{1}{2}LI_1^2,\ U_2 = \frac{1}{2}LI_2^2$

$\Delta U = \frac{1}{2}LI_2^2 - \frac{1}{2}LI_1^2$

$= \frac{1}{2} \times L\left(I_2^2 - I_1^2\right)$ ... (i)

$e = L\frac{di}{dt} \quad \therefore \quad 10 = \frac{L \cdot 5}{0.1} \quad \therefore \quad L = 0.2$ H

$\therefore \quad \Delta U = \frac{1}{2} \times \frac{2}{10}\left[10^2 - 5^2\right] = \frac{75}{10} = 7.5$ J

(32) Answer : (2)

Solution:

$X_L = \omega L$

$I = \frac{V_{\text{RMS}}}{Z} = \frac{V_{\text{RMS}}}{\sqrt{R^2 + X_L^2}}$ for *R-L* circuit

But $I = \frac{V_{\text{RMS}}}{\sqrt{R^2 + X_C^2}},\ X_C = \frac{1}{\omega C}$ for *R-C* circuit

As frequency increases, X_C decreases and current increases.

(33) Answer : (2)

Solution:

$V = V_0 \sin\omega t$

$V = 100\sqrt{2} \times \sqrt{2}\sin\left(2\pi \times 50t\right)$

$= 200 \sin(100\pi t)$

$= 200 \times \sin\left(100\pi \times \frac{1}{600}\right)$

$= 200\sin\left(\frac{\pi}{6}\right) = 100$ volt

(34) Answer : (4)

Solution:

Shift = 5 (fringe width)

$\frac{\left(\mu - 1\right)tD}{d} = \frac{5\lambda D}{d}$

$t = \frac{5\lambda}{\mu - 1} = 50000$ Å

(35) Answer : (4)

Solution:

Use Malus law, $I = I_0\cos^2\theta$

(36) Answer : (2)

Solution:

$t_{app} = \frac{t}{\mu}$

$t = \mu t_{app}.$

$t_1 = \mu \times 4$ cm

$t_2 = \mu \times 6$ cm

$t_1 + t_2 = \mu(4 + 6)$

$= 1.4(10) = 14$ cm

(37) Answer : (3)

Solution:

$\frac{1}{f'} = \left[\frac{\frac{3}{2}}{\frac{4}{3}} - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

(38) Answer : (2)

Solution:

Angular dispersion = $A(\mu_V - \mu_r)$

Angular dispersion = 6°(1.54–1.52)

= 6° × 0.02 = 0.12°

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(39) Answer : (2)
Solution:
Refractive index
$n = \sqrt{\mu_r \varepsilon_r}$
$= \sqrt{\frac{3}{16} \times 16}$
= 1.732

(40) Answer : (2)
Solution:
Energy of incident radiation = 4.2+2.0=6.2 eV
Wavelength of radiation,
$\lambda = \frac{12400}{6.2} = 2000\text{Å}$ (ultraviolet)

(41) Answer : (1)
Solution:
$h\upsilon = E_2 - E_1$

(42) Answer : (1)
Solution:
Mass number change only by α particle and atomic number by both α and β.
Since ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$
and ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}_{-1}e^{0}$
Hence ${}_{92}U^{238} \xrightarrow{9\alpha \text{ and } 8\beta^-} {}_{82}Pb^{202}$

(43) Answer : (3)
Solution:
Both P-type and N-type semiconductor are electrically neutral.

(44) Answer : (2)
Solution:
R_L = 1 kΩ
$l_L = \frac{8}{1} = 8\ \text{mA}$
$I_{500\Omega} = \frac{20-8}{500} = 24\ \text{mA}$
I_{diode} = 24 – 8 = 16 mA

(45) Answer : (1)
Solution:
In absence of an applied voltage, net current is zero.

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CHEMISTRY

(46) Answer : (1)
Hint:
Equivalents of $H_2C_2O_4 \cdot 2H_2O$ = Equivalents of NaOH
Solution:
Molarity of $H_2C_2O_4 \cdot 2H_2O = \frac{12.6 \times 1000}{126 \times 1000}$ = 0.1 M
Normality of $H_2C_2O_4 \cdot 2H_2O$ = 0.2
0.2 × 10 = 0.1 × V
V = 20 mL

$$\frac{\frac{(W)_{N_2}}{(MM)_{N_2}}}{\frac{(W)_{O_2}}{(MM)_{O_2}}} = \frac{(n)_{N_2}}{(n)_{O_2}}$$

$$\frac{(n)_{N_2}}{(n)_{O_2}} = \frac{\frac{1}{28}}{\frac{2}{32}} \Rightarrow \frac{16}{28} \Rightarrow \frac{4}{7}$$

(47) **Answer :** (3)
Hint:
O_2 is limiting reagent
Solution:

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$$

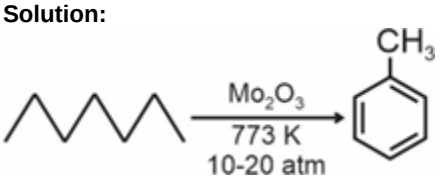
Mole 2 2

$\frac{2}{4}$ $\frac{2}{5}$

So, O_2 is limiting reagent
5 mole of O_2(g) produce → 4 mole of NO(g)
2 mole of O_2(g) produce → $\frac{4}{5} \times 2 = \frac{8}{5}$ = 1.6 mol of NO(g)
5 mole of O_2(g) produce → 6 mole of H_2O(g)
2 mole of O_2(g) produce → $\frac{6}{5} \times 2$ = 2.4 mole of H_2O(g)

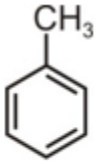
(48) **Answer :** (1)
Solution:
For H atom; Z = 1
For Li atom, Z = 3
Hence, E_{2s}(H) > E_{2s}(Li)

(49) **Answer :** (1)
Solution:

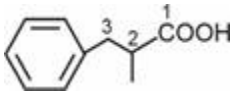


(50) **Answer :** (3)
Solution:
Stability of carbocation increases with adding of each cyclopropyl group due to conjugation between the bent orbitals of the cyclopropyl rings and the vacant p orbital of the cationic carbon.

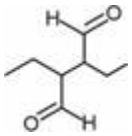
(51) **Answer :** (4)
Hint:
No double bond is possible at bridge head position
Solution:

-  shows hyperconjugation due to 3α hydrogen
- In carbanion, hyperconjugation is not possible.

(52) **Answer :** (4)
Solution:



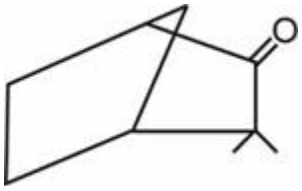
2-Methyl-3-phenylpropanoic acid

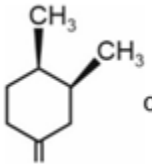
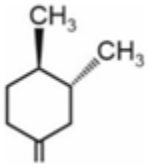


2,3-Diethylbutan-1,4-dial

$\overset{6}{HC} \equiv \overset{5}{C} - \overset{4}{CH_2} - \overset{3}{CH_2} - \overset{2}{CH} = \overset{1}{CH_2}$ Hex-1-en-5-yne

(53) Answer : (3)
Hint:
α-H from bridge head carbon should not be used for tautomerism since bridge head position cannot be sp^2 hybridised.
Solution:


contains one α-H but it is present at bridge head carbon.


cis isomer and

trans isomer

(54) Answer : (1)
Solution:
Lead sulphide is black in colour.

(55) Answer : (2)
Solution:
Gram equivalent of $KMnO_4$
= Gram equivalent of $Fe(NO_2)_2$
Moles × n = Moles × 5
Moles × 5 = 1 × 5
Moles = $\frac{5}{5}$
= 1

(56) Answer : (1)
Hint:
$[OH^-] = 10^{-pOH}$
Solution:
pOH = 14 – 11 = 3 ⇒ $[OH^-] = 10^{-3}$

$$X(OH)_2(s) \rightleftharpoons X^{2+}(aq) + 2OH^-(aq)$$
$$s \qquad 2s = 10^{-3}$$

At equilibrium, $K_{sp} = [X^{2+}][OH^-]^2 = \left(\frac{1}{2} \times 10^{-3}\right)(10^{-3})^2$
$K_{sp} = 5 \times 10^{-10}$

(57) Answer : (1)
Hint:
Mixture of CH_3COOH and CH_3COONa will act as buffer solution.
Solution:
For 1st solution:

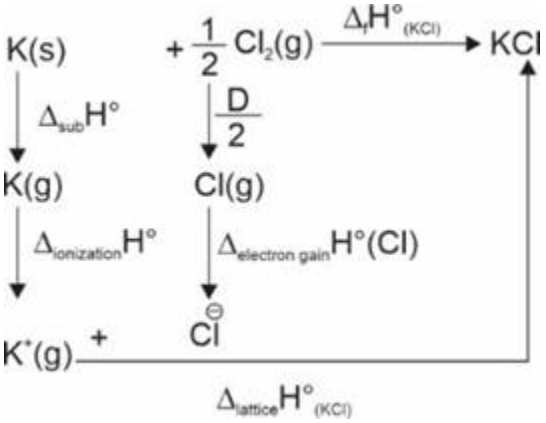
$$CH_3COONa + HCl \longrightarrow CH_3COOH + NaCl$$

t = 0	1M	1M	0
t = t	0	0	1M

$[H^+] = \sqrt{K_a \times C}$
$pH = \frac{1}{2}pK_a = a$
For 2nd solution:
$b = pK_a + \log\frac{[CH_3COONa]}{[CH_3COOH]}$
$b = pK_a$
$a : b = \frac{pK_a}{2 \times pK_a} = 1 : 2$

(58) Answer : (2)
Solution:

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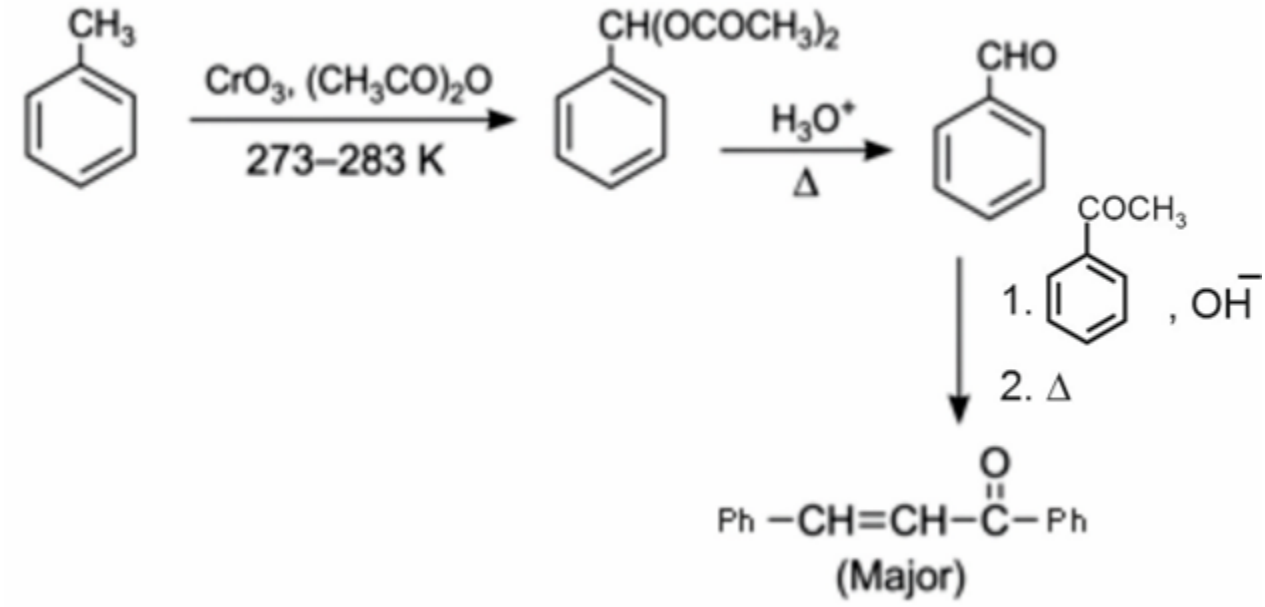


$\Delta_f H^\circ(KCl) = \Delta_{sub}H^\circ_{(K)} + \Delta^\circ_{ionization} + H^\circ(K) + \frac{1}{2}\Delta_{bond}H^\circ(Cl_2) + \Delta_{electron\ gain}H^\circ(Cl) + \Delta_{lattice}H^\circ_{(KCl)}$

$-436.7 = 89.2 + 419 + \frac{1}{2}(243.0) + x - 717.8$

$x = -348.6\ kJ\ mol^{-1}$

(59) Answer : (4)
Solution:



(60) Answer : (3)
Solution:

$$O_2 \xrightarrow{+2e^-} O_2^{2-}$$
$$B.O \rightarrow 2 \qquad 1$$

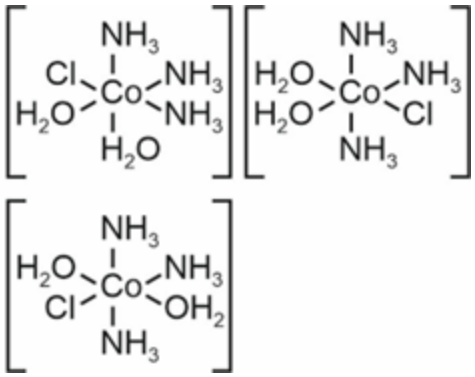
Bond order decreases, bond length increases.

The correct order of bond angle is $\overset{+}{N}O_2 > \dot{N}O_2 > NO_2^-$

Molecule	Bond Length (pm)
H_2	74.2
Li_2	267.2
B_2	158.9
F_2	141.8

(61) Answer : (3)
Solution:
BF_3 is trigonal planar in shape and has 120° bond angle.

(62) Answer : (4)
Solution:

$[Co(NH_3)_3(H_2O)_2Cl]Cl_2 \Rightarrow$ 

$\Delta T_f = i\, k_f m$
1.116 = i × 1.86 × 0.2
i = 3
so, the complex (A) will be $[Co(NH_3)_5Cl]Cl_2$.
This complex has zero unpaired electron so its magnetic moment is zero.
• If one NH_3 ligand is replaced by H_2O molecule then complex becomes $[Co(NH_3)_4(H_2O)Cl]Cl_2$ which does not show optical isomerism.
• If two NH_3 ligand are replaced by two H_2O molecule then complex becomes $[Co(NH_3)_3(H_2O)_2Cl]Cl_2$ which has three stereoisomers.

(63) Answer : (3)
Solution:
The correct order of first ionisation energy of group 13 element is
Al < Ga < Tl < B

Elements	$\Delta_{eg}H$
H	– 73
Li	– 60
Na	– 53
K	– 48

(64) Answer : (1)
Hint:
Osmotic pressure π = CRT
Solution:
As solute is non-electrolyte, so *i* = 1 taking R = 0.082 L-bar mol^{-1} K^{-1}
So, slope of the straight line is RT
26.24 = 0.082 T
∴ T = 320 K
= 320 – 273 = 47°C

(65) Answer : (1)
Solution:
For pure a, $X_A = 1$
$P_A^0 = 100 + 200 = 300$ mm of Hg
For pure B, $X_B = 1$, $X_A = 0$
$P_B^0 = 100$ mm of Hg

(66) Answer : (1)
Hint:
1 F charge deposits 1 gram equivalent metal during electrolysis.
Solution:
1 mol Cu from $CuSO_4$ is 2 eq. Cu
Hence 2 F required.
1 mol Al from $AlCl_3$(l) is 3 eq. Al
Hence 3 F required.
1 mol Ag from AgBr is 1 eq.
Hence 1 F required.

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1 mol Fe from Fe_3O_4 is $\frac{8}{3}$ eq.

Hence $\frac{8}{3}$ F required.

(67) **Answer :** (4)
Solution:

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ $\quad \Delta G_1^0 = -Fx_1$ (i)

$Fe^{3+} + 3e^- \rightarrow Fe$ $\quad \Delta G_2^0 = -3Fx_2$ (ii)

(ii) − (i)

$Fe^{2+} + 2e^- \rightarrow Fe$

$\Delta G_3^0 = -2\,Fx_3$

$-2Fx_3 = Fx_1 - 3Fx_2$

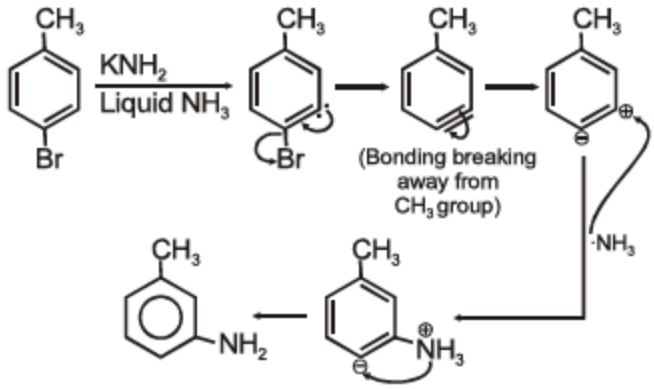
$x_3 = \frac{3x_2 - x_1}{2}$

(68) **Answer :** (1)
Solution:
For first order reaction

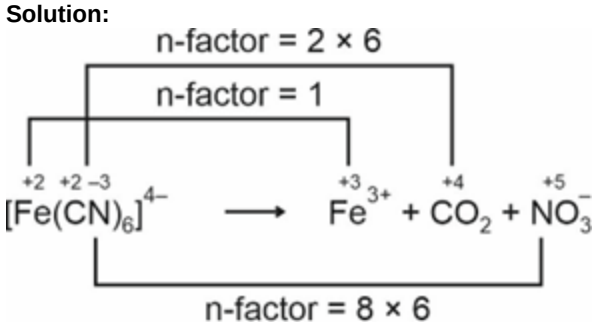
$k = \frac{2.303}{60} \log \frac{100}{40} = 0.0153\ min^{-1}$

$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.0153} = 45.3$ min

(69) **Answer :** (2)
Solution:
Answer (2)

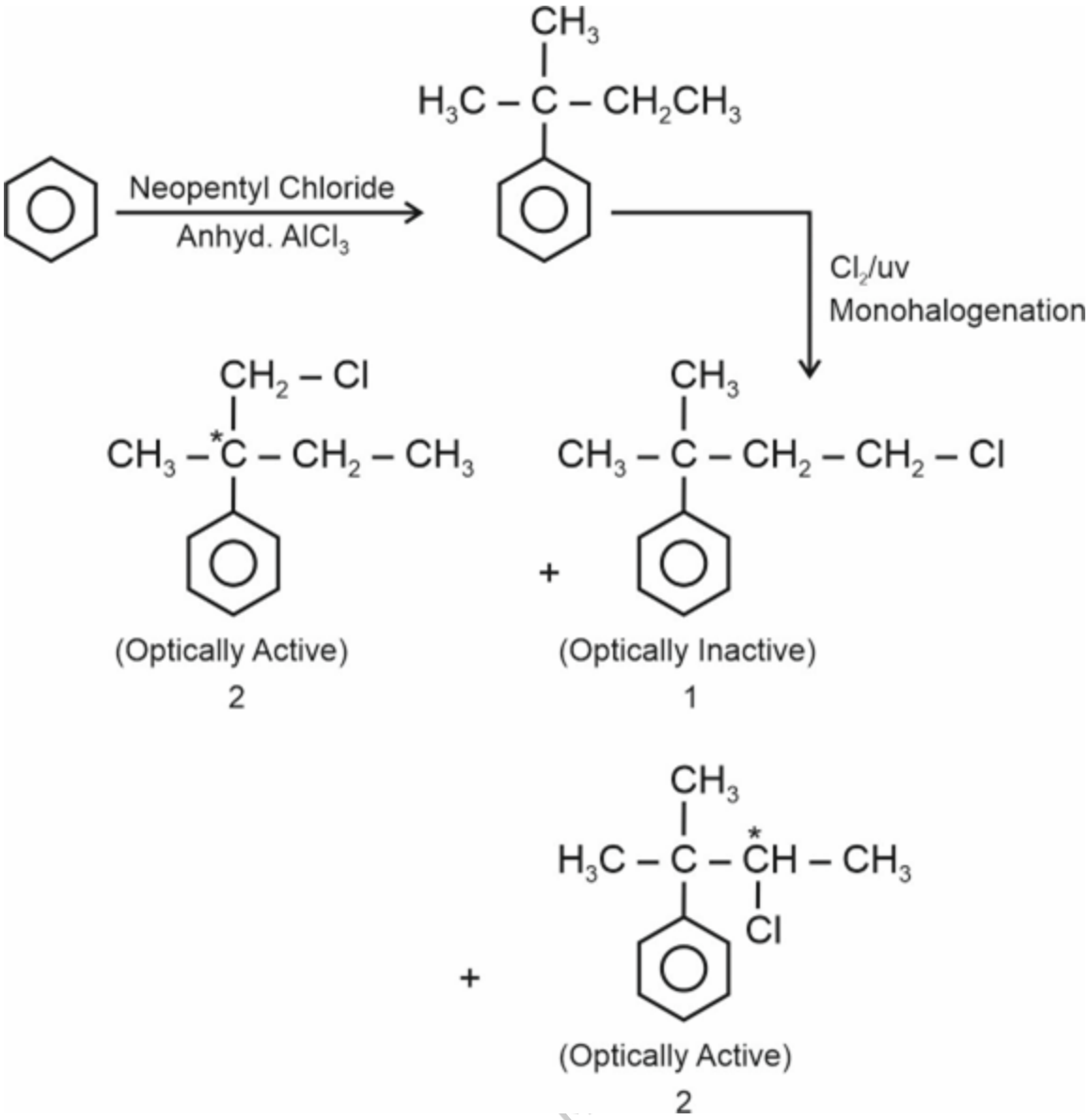


(70) **Answer :** (2)
Solution:



Total n-factor = 48 + 12 + 1 = 61

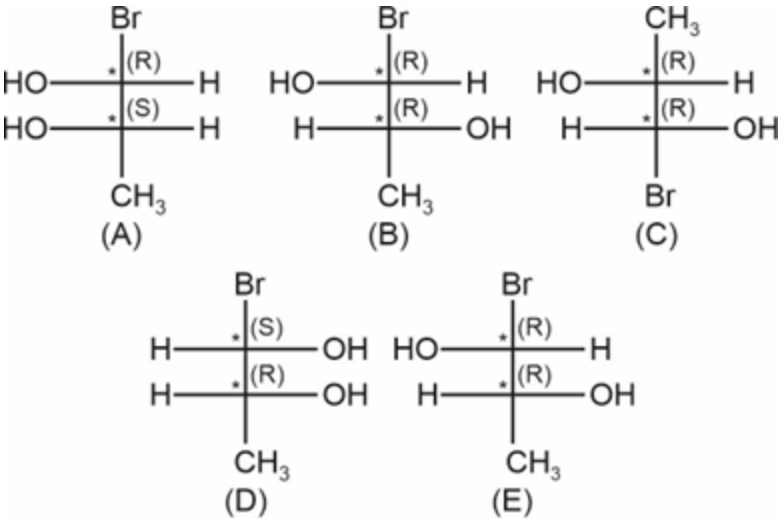
(71) **Answer :** (4)
Solution:



Total 5 isomers are possible.

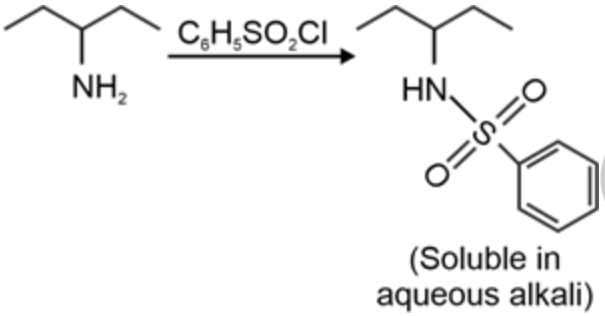
(72) **Answer :** (1)
Solution:
Amylopectin is branched chain polymer of α-D-glucose units in which chain is formed by C1 – C4 glycosidic linkage whereas branching occurs by C1 – C6 glycosidic linkage.

(73) **Answer :** (2)
Solution:



* Diastereomer	Identical	Enantiomers
A & B, A & E C & D, D & E	B & C, A & E A & C	A & D

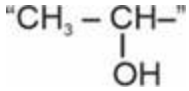
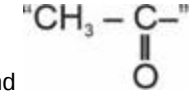
(74) Answer : (3)
Solution:

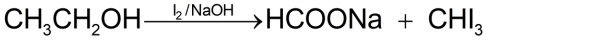


(Soluble in aqueous alkali)

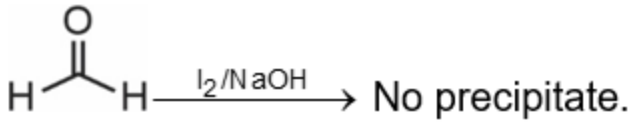
(75) Answer : (3)
Solution:
Aliphatic and aromatic primary amines on heating with chloroform and ethanolic KOH form isocyanide.

(76) Answer : (2)
Hint:
During cannizzaro reaction, disproportionation of aldehydes takes place with conc. NaOH.
Solution:
Aldehydes which do not have an α-hydrogen atom, undergo oxidation and reduction (disproportionation) reaction on heating with concentrated NaOH or KOH.
For example : Formaldehyde, benzaldehyde.

(77) Answer : (4)
Hint:
Species with  and  groups can give Haloform reaction.
Solution:

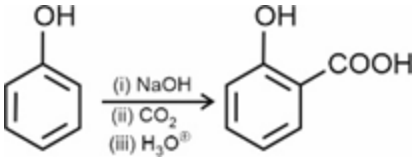


yellow precipitate

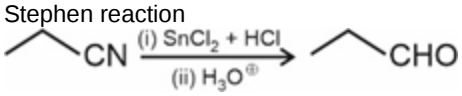


(78) Answer : (4)
Solution:

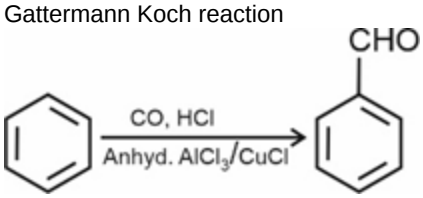
Kolbe's reaction



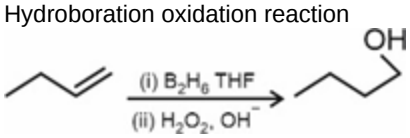
Stephen reaction



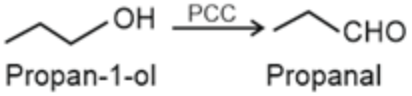
Gattermann Koch reaction



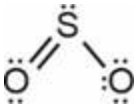
Hydroboration oxidation reaction



(79) **Answer :** (1)
Solution:



(80) **Answer :** (3)
Hint:
Structure of SO_2 is



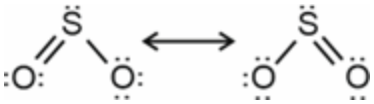
Solution:

$$SO_2(g) + Cl_2(g) \xrightarrow{\text{Charcoal}} SO_2Cl_2(l)\ \left(\text{sulphuryl chloride}\right)$$

• $2Fe^{3+} + SO_2 + 2H_2O \rightarrow 2Fe^{2+} + SO_4^{2-} + 4H^+$

Moist SO_2 acts as reducing agent

•



It is a resonance hybrid of two canonical forms.

(81) **Answer :** (2)
Solution:

$$\begin{array}{lccccc} & X(g) & \rightarrow & 2Y(g) & + & Z(g) \\ t=0 & P_0 & & - & & - \\ t=t & P_0-P & & 2P & & P \end{array}$$

$t = \frac{2.303}{k}\log\frac{2P_0}{2P_0-P_t+P_0}$

$t = \frac{1}{k}\ln 2P_0 - \left(\ln 3P_0 - P_t\right)$

$\ln(3P_0 - P_t) = \ln 2P_0 - P_t$

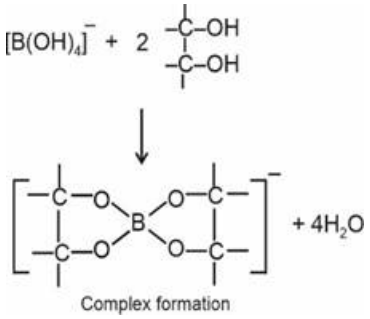
$y = c - mx$

At any time 't', rate constant is independent of concentration $[X]_0$.

(82) **Answer :** (4)
Solution:

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H_3BO_3 does not undergo self-ionization.



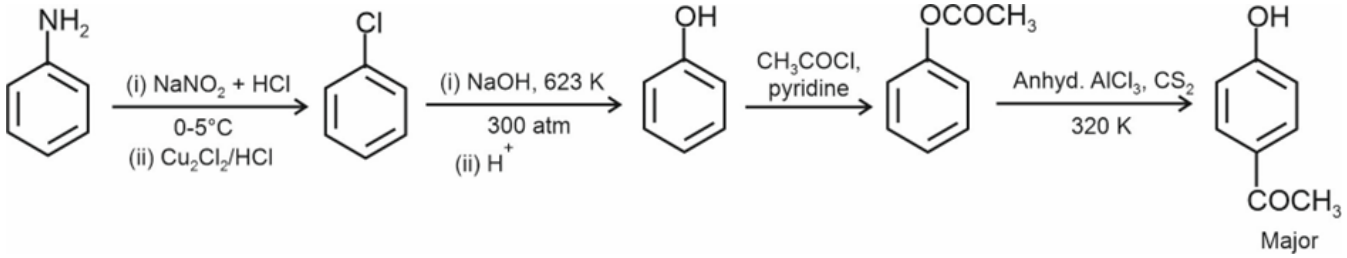
Complex formation

(83) Answer : (1)
Solution:
Both $Ce^{4+}(4f^0)$ and $Lu^{3+}(4f^{14})$ are diamagnetic.

(84) Answer : (4)
Hint:
More is the electron density on metal, more will be the back donation from metal to π* orbital of CO
Solution:
More is the back donation from metal to the π* orbital of CO, stronger is M–C bond and weaker is C–O bond strength. Correct order of C–O bond strength:
$[Mn(CO)_6]^+ > [Cr(CO)_6] > [V(CO)_6]^-$

(85) Answer : (2)
Solution:
Process A is adiabatic as entropy is constant.
$\frac{\Delta U}{R} = \frac{U_f}{R} - \frac{U_i}{R}$
= 1300 – 2650
$\Delta U = \omega = -1350R$
$\Delta U = nC_v \Delta T$
$-1350R = 1 \times \frac{5}{2}R \times \left(T_2 - T_1\right)$
$T_2 - 900 = -540$
$T_2 = 360K$
Process 'B' is isothermal as $\frac{U}{R}$ is constant
$\omega_B = -nRT_2 \ln\left(\frac{V_3}{V_2}\right)$
$-1350R = -1 \times R \times 360 \ln \frac{V_3}{V_2}$
$-1350R = -360\,R\ln\left(\frac{V_3}{V_2}\right)$
$\ln\left(\frac{V_3}{V_2}\right) = 3.75$

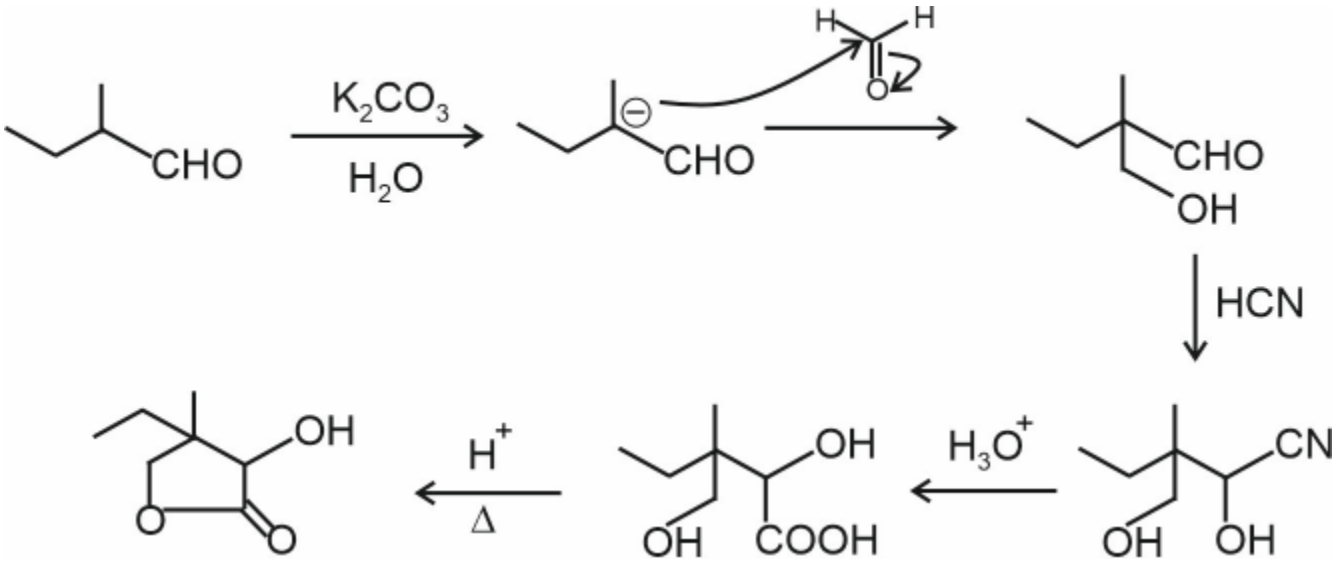
(86) Answer : (2)
Solution:



Major

(87) Answer : (4)
Solution:

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(88) Answer : (4)
Solution:

The total number of spectral lines = $\frac{\Delta n(\Delta n+1)}{2}$

For (a), total number of spectral line = $\frac{2\times 3}{2} = 3$

For (b), total number of spectral lines = $\frac{3\times 4}{2} = 6$

(89) Answer : (4)
Hint:
Group reagents are used for the separation of cation due to their selective precipitation.
Solution:

Group	Group resent
Group-I	Dilute HCl
Group-II	H_2S gas in presence of dil.HCl
Group-III	NH_4OH in presence of NH_4Cl
Group-IV	H_2S in presence of NH_4OH

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(90) Answer : (3)
Solution:
2° alcohols give blue colour in Victor Meyer's test.

BIOLOGY

(91) Answer : (2)
Solution:
A – The family of housefly is Muscidae
B – The class of wheat is Monocotyledonae
C – The order of man is Primata
D – The genus of mango is *Mangifera*
So, D, A, B and C respectively are *Mangifera*, Muscidae, Monocotyledonae and Primata.

(92) Answer : (4)
Hint:
Cuscuta is a parasitic plant.
Solution:
In imperfect fungi, reproduction is only by asexual spores known as conidia.

(93) Answer : (4)
Hint:
This organism is *Nostoc.*
Solution:

Nostoc is a cyanobacterium that can fix both atmospheric nitrogen and carbon dioxide.

(94) Answer : (4)
Solution:
Sleeping sickness causing organism is *Trypanosoma.*

(95) Answer : (2)
Solution:
Male strobili produces only one type of spores. In *Cycas,* male cones and megasporophylls are borne on different trees.

(96) Answer : (1)
Solution:
Gymnosperms are archegoniate phanerogams while angiosperms are non-archegoniate phanerogams.
Pinus and *Cycas* – Produce seeds and have archegonia.
Funaria, *Selaginella* – Have archegonia but lack seeds.
Mustard – Have seeds but lack archegonia.

(97) Answer : (1)
Solution:
Chemotaxonomy is based on biochemical component of a cell.

(98) Answer : (3)
Hint:
Basal placentation is found in marigold.
Solution:

Plant Name	Type of placentation
(a) *Argemone*	• Parietal
(b) *Marigold*	• Basal
(c) *Dianthus*	• Free-central
(d) Lemon	• Axile

(99) Answer : (3)
Solution:
A bud is present in the axil of the simple and compound leaf but is not present in the axil of each leaflet.
Both simple and compound leaves can have reticulate venation. A compound leaf's lamina is divided into many leaflets when the incision touches the midrib.

(100) Answer : (1)
Solution:
Seeds of coconut, onion, maize and barley are endospermous.

(101) Answer : (1)
Solution:
Epiblema is the root epidermis which gives rise to the root hair which help to absorb water and minerals from the soil.

(102) Answer : (3)
Hint:
Label A is SER, Label B is RER, Label C is nuclear pore and Label D is ribosome.
Solution:
RER is extensive and continuous with the outer membrane of the nucleus. SER synthesises lipid-like steroidal hormones in animal cells. Nuclear pore provides the passage through which movement of RNA and protein molecules takes place in both the directions between the nucleus and the cytoplasm. Ribosomes are ribonucleoproteins.

(103) Answer : (3)
Solution:
In dicot stems, pericycle forms semi-lunar patches of sclerenchymatous cells.

(104) Answer : (3)
Solution:
The cell theory given by Schleiden and Schwann did not explain how new cells were formed.
Rudolf Virchow first explained that cells divided and new cells are formed from pre-existing cells, *i.e*., *Omnis cellula-e cellula*.

(105) Answer : (4)
Hint:
Schwann proposed hypothesis that the bodies of animals and plants are composed of cell and product of cell.
Solution:
Theodore Schwann studied different type of animal cells and reported that cell had a thin outer layer.

(106) Answer : (4)

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Solution:
Squamous epithelium forms the innermost layer of walls of blood vessels (endothelium), air sac (alveoli) of the lungs.

(107) Answer : (1)
Solution:
CO_2 can diffuse through the diffusion membrane more rapidly than O_2 per unit difference in partial pressure as solubility of CO_2 in plasma is 20-25 times higher than that of O_2.

(108) Answer : (3)
Solution:
Primary metabolites such as amino acids and sugars have identifiable functions in normal physiological processes.

(109) Answer : (4)
Solution:

Element	% Weight of Earth's crust	% Weight of Human body
Hydrogen (H)	0.14	9. 5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	Very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium(Ca)	3.6	1.5
Magnesium(Mg)	2.1	0.1
Silicon (Si)	27.7	Negligible

(110) Answer : (4)
Solution:
• The chemical and physical properties of amino acids are essentially of the amino, carboxyl and the R functional groups.
• In serine, R-functional group is hydroxy methyl.

(111) Answer : (2)
Solution:
When two or more polypeptide subunits come together, it is called quaternary structure. Alpha helix or beta-pleated sheet represent secondary structures of proteins.

(112) Answer : (3)
Solution:
Arachidonic acid has 20 carbon atoms including the carboxyl carbon. Stearic acid has 18 carbon atoms and Palmitic acid has 16 carbon atoms.

(113) Answer : (4)
Solution:
Muscular diaphragm and mammary glands are found in all mammals. Some mammals have homodont dentition. Some mammals have intra-abdominal testes.

(114) Answer : (3)
Solution:
Due to the absence of air bladder, cartilaginous fishes have to swim constantly to avoid sinking.

(115) Answer : (4)
Solution:
In chordates, notochord is present ventral to nerve cord.

(116) Answer : (1)
Solution:
Mycoplasma is the smallest living cell. It is a prokaryotic organism.

(117) Answer : (1)
Solution:
In contrast to meiosis I, meiosis II resembles a normal mitosis.
In metaphase II, chromosome align at the equator and microtubules get attached to the kinetochores of sister chromatids.

(118) Answer : (2)
Hint:

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DNA replication takes place only once during meiosis.

Solution:

Meiosis involves two sequential cycles of nuclear and cell division but only a single cycle of DNA replication.

(119) Answer : (3)

Hint:

The sequence of phases in M phase is karyokinesis and then cytokinesis.

Solution:

The accurate sequence of the events in M phase is as follows:

b. Prophase → Centrosome begins to move towards opposite poles of the cell.

a. Metaphase → Spindle fibres attach to kinetochores of chromosomes.

c. Anaphase → Chromosomes begin their migration towards the two opposite poles.

e. Telophase → Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

d. Cytokinesis → Appearance of a furrow in the plasma membrane.

(120) Answer : (3)

Solution:

The products of light reaction of photosynthesis are ATP, NADPH and oxygen.

(121) Answer : (1)

Solution:

Sorghum is a C_4 plant.

Sorghum, being a C_4 plant has Kranz anatomy in their leaves.

(122) Answer : (3)

Solution:

During Krebs cycle, substrate level phosphorylation occurs during conversion of succinyl CoA to succinic acid.

(123) Answer : (4)

Solution:

$$\text{R.Q.} = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$$

It is more than one for organic acids.

(124) Answer : (4)

Solution:

The given graphical representation is of geometric growth. In the graph, A represents lag phase of growth.

(125) Answer : (2)

Solution:

Gibberellins induce internode elongation just prior to flowering in beet and cabbage.

(126) Answer : (4)

Solution:

Polyembryony is occurrence of more than one embryo in a seed. In false fruit, thalamus also contributes to fruit formation.

Mangifera belongs to family Anacardiaceae.

(127) Answer : (4)

Hint:

Radical and root cap in monocot embryo are enclosed in sheath.

Solution:

In monocots, the lower ends of the embryonal axis has radical and root cap enclosed in an undifferentiated sheath called coleorhiza.

(128) Answer : (4)

Solution:

The primary endosperm nucleus is produced by the fusion of a male gamete and two fused polar nuclei. This process is known as triple fusion, resulting in a triploid nucleus.

(129) Answer : (1)

Solution:

The females in F_1 generation would be heterozygous normal for both colour blindness and phenylketonuria.

50% of the male children will suffer with colorblindness and 50% with phenylketonuria.

(130) Answer : (1)

Solution:

The alignment of genes on chromosome can be adcb or adbc.

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(131) Answer : (2)
Solution:
Each segment of a cockroach's body is covered by 4 sclerites *i.e*., dorsal tergite (1), ventral sternite (1) and lateral pleurite (2).

(132) Answer : (4)
Solution:
Blood has proteins, whereas very few proteins are present in lymph. The RBCs of frogs are nucleated.

(133) Answer : (2)
Solution:
Compound epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.

(134) Answer : (3)
Solution:
When the oxygen-haemoglobin dissociation curve shifts to the right side, it generally indicates a decreased affinity of haemoglobin for oxygen, which facilitates the dissociation of oxygen from haemoglobin. This shift is typically associated with conditions such as increased temperature, increased levels of CO_2, or decreased pH (Bohr's effect). As a result, there is a less percentage saturation of haemoglobin at a given partial pressure of oxygen. However, the P_{50} value, which is the partial pressure of oxygen at which haemoglobin is 50% saturated, actually increases when the curve shifts to the right.

(135) Answer : (3)
Hint:
Heart chamber which has thickest myocardium.
Solution:
Left ventricle pumps blood to all systems of the body by its forceful contraction. So, its wall has thicker musculature in comparison with any other chamber of the heart.

(136) Answer : (1)
Solution:
Erythroblastosis foetalis occurs if Rh-ve mother delivers Rh +ve baby in subsequent pregnancy with Rh+ve baby.

(137) Answer : (3)
Solution:
SA node is called pacemaker and it is present at top right corner of right atrium.

(138) Answer : (3)
Solution:
Urinary bladder and urethra are unpaired.

(139) Answer : (1)
Solution:
When the osmolarity of the blood increases, this stimulates the hypothalamus to release ADH.
Renin is released from JG cells of kidneys. Aldosterone is released from adrenal cortex.

(140) Answer : (4)
Solution:
Troponin and tropomyosin are regulatory proteins in skeletal muscle.

(141) Answer : (3)
Solution:
Hint: Filter feeding
Sol.: *Hydra* can use its tentacles for capturing its prey and also use them for locomotion. *Amoeba* forms pseudopodia by the streaming of protoplasm. Ciliary movement in *Paramoecium* helps in the movement of food through cytopharynx and in locomotion as well.

(142) Answer : (1)
Solution:
The neural system of all animals is composed of highly specialised cells called neurons which can detect, receive and transmit different kinds of stimuli. When a motor neuron is suitably stimulated, the influx of Na^+ increases due to increase in permeability of Na^+ that leads to depolarisation.

(143) Answer : (3)
Solution:
The trophoectoderm is the outer layer of cells in the blastocyst stage, not the morula. It eventually contributes to the formation of the placenta.
Colostrum, the first form of milk produced after childbirth, is rich in IgA which plays a crucial role in providing passive immunity to the newborn.

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In humans, implantation of the blastocyst into the uterine wall typically occurs about 6 to 10 days after fertilization, not after one month.

(144) Answer : (4)
Solution:
Relaxin, a hormone produced first by the corpus luteum of the ovary and later by the placenta, increases the flexibility of the pubic symphysis by softening of ligaments and helps in dilation of cervix during labor.

(145) Answer : (2)
Solution:
The seminiferous tubules of the testis open into the vasa efferentia through rete testis. The vasa efferentia leave the testis and open into epididymis located along the posterior surface of each testis. The epididymis leads to vas deferens that ascends to the abdomen and loops over the urinary bladder.

(146) Answer : (3)
Solution:
Condoms are not reusable and CuT and implants cannot be used without the help of any medical practitioner.

(147) Answer : (2)
Solution:
Venereal diseases are those diseases or infections that spread through sexual intercourse.

(148) Answer : (4)
Hint:
A single huge explosion formed Universe.
Solution:
Origin of species was explained by Charles Darwin and Origin of life by A.I. Oparin. Origin of Universe is explained by Big-Bang hypothesis.

(149) Answer : (3)
Solution:
First vertebrates were cyclostomes while earliest gnathostomes were Pisces. Coelacanth has bony endoskeleton.

(150) Answer : (2)
Solution:
Ramapithecus was more man-like while *Dryopithecus* was more ape-like. *Homo habilis* probably did not eat meat. *Homo erectus* probably ate meat.

(151) Answer : (3)
Solution:
Physiological barriers are tears in eyes, saliva in mouth, HCl in stomach, *etc*.

(152) Answer : (1)
Solution:
Bacteria such as *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for the disease pneumonia.

(153) Answer : (3)
Solution:
Restriction enzymes are called as “The Molecular Scissors”. The second component is a modification enzyme that adds a methyl group to one or two bases in a DNA sequence recognised by the restriction enzyme. Once a base in a DNA sequence is modified by the addition of a methyl group, the restriction enzymes fail to recognise and cleave that DNA.

(154) Answer : (4)
Solution:
Denaturation in PCR – Requires temperature above 90°C.

(155) Answer : (3)
Solution:
The letter ‘R’ in the name of restriction endonuclease *Eco* RI denotes the strain of the source organism.

(156) Answer : (2)
Solution:

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Parents: BBYY x bbyy

F_1: BbYy

Selfing of F_1: BbYy × BbYy

F_2:

♀ \ ♂	BY	By	bY	by
BY	BBYY	BBYy	BbYY	BbYy
By	BBYy	BByy	BbYy	Bbyy
bY	BbYY	BbYy	bbY Y	bbYy
by	BbYy	Bbyy	bbYy	bbyy

The phenotypic ratio regarding starch grain size and seed colour would be 3 : 6 : 3 : 1 : 2 : 1

(157) Answer : (1)

Solution:

Human females have 44 autosomes and two X chromosomes.

(158) Answer : (3)

Solution:

Lambda phage has dsDNA with 48502 bp. UTRs are present at 5'-end before start codon and 3'-end after stop codon.
Maurice Wilkins and Rosalind Franklin had not proposed a definitive model for DNA.

(159) Answer : (4)

Solution:

Total segment length = 51 nm
Length of one complete turn of DNA = 3.4 nm
Hence, number of turns in the DNA segment
= $\frac{51}{3.4}$ = 15 turns
Total base pairs = 150; total number of bases = 300
Adenine is 20% of 300, therefore
$\frac{20}{100} \times 300 = 60$ adenine
By Chargaff's equivalency we know,
Adenine = Thymine = 60
So AT pairs = 60
And GC pairs = 90
We know AT pairs contains 2 hydrogen bonds each whereas GC pairs contain 3 hydrogen bonds.
So, total hydrogen bonds = (60 × 2) + (90 × 3)
= 120 + 270 = 390

(160) Answer : (3)

Solution:

Due to semiconservative DNA replication, after 3 generations 8 daughter cells would be formed out of which 6 will show no-radioactivity.

(161) Answer : (3)

Solution:

An mRNA has many non-coding stretches of nucleotides and the codons are always triplet. Therefore, if there are n number of nucleotides in an mRNA, the maximum number of amino acids in the polypeptide synthesised from this mRNA would be approximately $\frac{n}{3}$.

(162) Answer : (2)

Solution:

The *lac* y gene codes for permease.

(163) Answer : (2)
Solution:
An example of microbial biocontrol agents that can be introduced in order to control butterfly caterpillars is the bacteria *Bacillus thuringiensis*.

(164) Answer : (1)
Solution:
Acetobacter aceti yields acetic acid and it is used in preparation of vinegar. *Aspergillus niger* helps in production of citric acid used for preserving food.

(165) Answer : (4)
Solution:
Population attains asymptote when population size (N) approaches carrying capacity K. The rate of a population approaching asymptote depends on *r* (intrinsic rate of natural increase). So, the population with highest 'r' reaches the asymptote the quickest.

(166) Answer : (2)
Solution:
In predation, parasitism and amensalism, one of the interacting species is harmed.
In competition, both the interacting species are harmed.

(167) Answer : (3)
Solution:
This pyramid cannot represent a tree ecosystem. The pyramid of number in tree ecosystem is either inverted or spindle shaped.

(168) Answer : (1)
Solution:
GFC is the major conduit of energy flow in aquatic ecosystem.

(169) Answer : (1)
Solution:
Hotspots show high level of species richness. There are three biodiversity hotspots in India.

(170) Answer : (2)
Solution:
The problem in identifying all the species of bacteria is that conventional taxonomic methods are not suitable for identifying microbial species and many species are simply not culturable under laboratory conditions. Biologists are not sure about how many prokaryotic species there might be.

(171) Answer : (1)
Solution:
The smaller the DNA fragment, the farther it moves.

(172) Answer : (4)
Solution:
A recognition site for a restriction endonuclease is typically a specific sequence of nucleotides that is palindromic, meaning it reads the same forwards and backwards on complementary strands. This is because restriction enzymes often cut DNA at these palindromic sequences.

(173) Answer : (3)
Solution:
For chemical safety testing, transgenic animals are made more sensitive to toxic substances.

(174) Answer : (4)
Solution:
In skeletal muscles, many nuclei occur at irregular intervals and are peripheral in position. Because these fibres are multinucleated, they form a structural syncytium, for *e.g.*, thigh muscles, biceps.

(175) Answer : (2)
Hint:
Identify a polymer of fructose.
Solution:
Inulin is obtained from the roots of plants like *Dahlia*. It is a polymer of fructose whereas starch, glycogen and cellulose are polymers of glucose.

(176) Answer : (3)
Solution:

Creation of different breeds of dogs is an example of artificial selection.

(177) Answer : (1)
Solution:
GIP inhibits gastric secretion and motility.

(178) Answer : (3)
Solution:
Glucagon is a proteinaceous hormone and produces second messenger when it combines with its membrane-bound receptor. Hormone-receptor complex activates G-protein to form 2nd messenger, which is cyclic AMP.

(179) Answer : (1)
Solution:
Patient 'X' is suffering from secondary hypothyroidism in which synthesis of T_4 and T_3 becomes impaired due to low TSH. When TSH is given by injection, synthesis of T_4 rises, it means thyroid gland was normal and only pituitary gland was responsible for hypothyroidism.

(180) Answer : (3)
Solution:
Sympathetic nervous system is stimulated during fight or flight response.

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